Safe Automation of Administrative Document Pre-Checks through Cross-Document Verification

Munira Olimjonova[1], **Khabibillo Gulomov**[1], Hyunji Lee[2], Dongkyun Kim[1,*]

[1]School of Computer Science and Engineering, Kyungpook National University [2]Flash21 Co., Ltd.
* Corresponding author khabibillo@knu.ac.kr

56 / 78
business submissions Ready with no staff review

6.9 → 0.63
items left for staff per business submission

8.19 → 3.50
items left for staff per student submission

1 Problem and goals

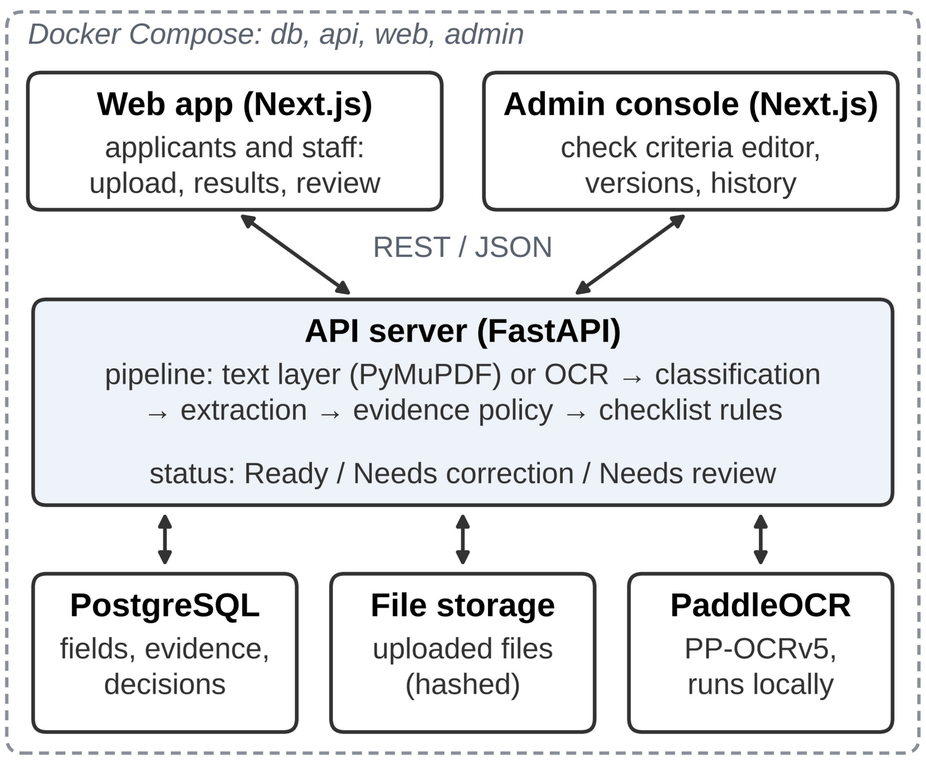
- Staff check grant and scholarship certificates by hand. An expired certificate or another company's document can slip into the decision.
- Goals: automate routine checks, never mark a submission Ready on a guess, let administrators change the rules without code, and explain every result.

2 How ADVS decides

ADVS reads each certificate (PDF text or OCR), extracts its fields and checks them against the rules. A value is accepted only when a separate source supports it; everything else goes to staff.

Value	Confirmed by
Business submission	
Business registration no.	tax and insurance certificates; check digit
National tax validity	issue date + 30 days (by law)
Student submission	
Name, birth date, issue date	same values on the transcript
GPA	GPA typed by the applicant

3 System architecture



Next.js · FastAPI · PostgreSQL · PaddleOCR (runs locally) · Docker Compose

4 Validation

- All 9 demonstration scenarios gave the expected result (complete, expired, mismatched, look-alike, missing, scanned).
- 889 back-end and 115 front-end automated tests.
- Real certificates of one university and two companies as scans, photos and files: invalid submissions were never Ready.

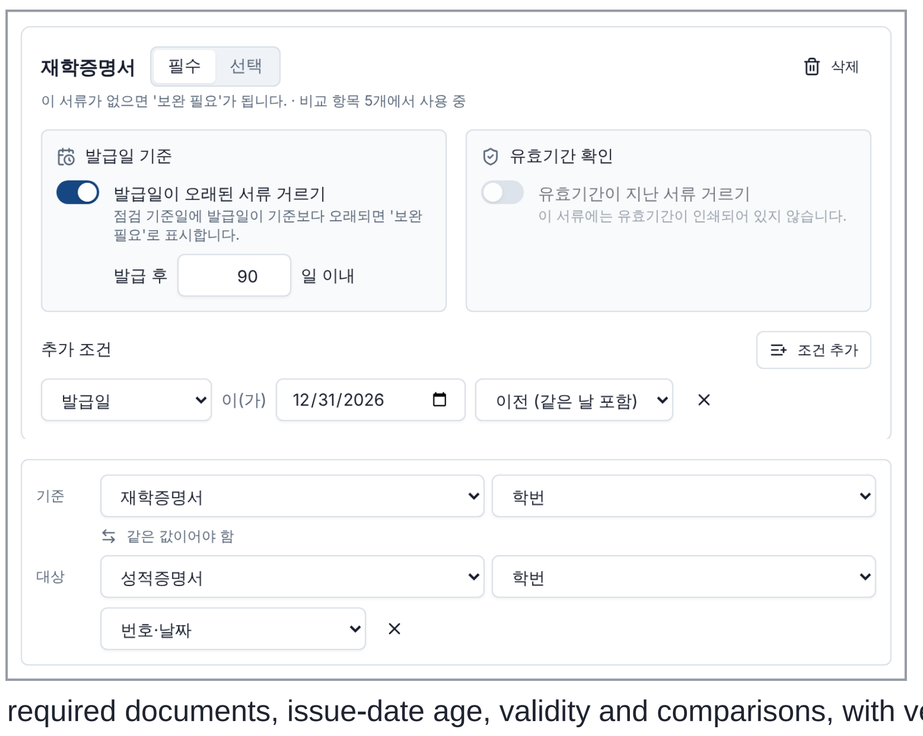
5 What users see



Applicant: status first, then each problem and how to fix it.



Staff: confirm unsupported values next to the original, one key press.



Admin: required documents, issue-date age, validity and comparisons, with versions.

6 Conclusion and next steps

ADVS automates routine document checks while keeping staff in control, and administrators adapt the rules without code. Next: staff accounts, electronic certificates, and more schools and companies.

Source code on GitHub

This research was supported by the MSIT, Korea, under the National Program for Excellence in SW (2021-0-01082) supervised by the IITP, and by the MSIT and NIPA through the Regional Industry SW Talent Development Infrastructure Program.